

Economies of Scale, Imperfect Competition, and International Trade

ECON 308 – International Economic Relations Chapter 7

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Opening Puzzle: Why Do Canada and Germany Both Export Cars?

- Germany and Canada are both capital-abundant, high-income countries with similar technology.
- H-O predicts *little* trade between them.
- **Reality:** Canada exports Chrysler minivans and Lincoln SUVs; Germany exports BMW and Mercedes.
- This is **intra-industry trade (IIT)**: simultaneously exporting and importing the same product category.
- The Ricardian and H-O models *cannot explain* IIT. We need a new framework.

New Insight Needed

When there are *economies of scale* and *product differentiation*, each country specialises in different *varieties* of a good — not different goods. Trade expands variety without eliminating production at home.

Learning Objectives

1. Distinguish external from internal economies of scale and their implications.
2. Derive the equilibrium in a monopolistically competitive industry.
3. Show how trade expands the market and generates gains via the Krugman model.
4. Measure intra-industry trade using the Grubel-Lloyd index.
5. Explain why industrial clusters arise and exhibit path dependence.
6. Evaluate the policy implications of economies of scale.

Economies of Scale: The Basic Concept

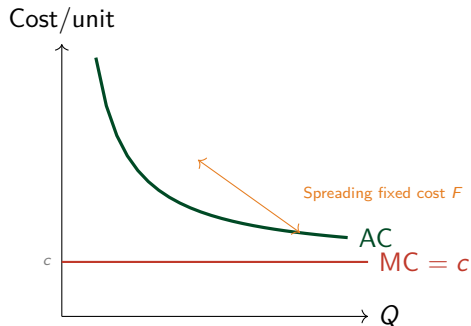
Economies of Scale (EoS)

Average cost *falls* as output rises:

$AC(Q) = F/Q + c$, where F is fixed cost and c is constant marginal cost.

Why EoS arise:

- Fixed costs (R&D, tooling, brand) spread over more units.
- Specialisation of labour and equipment at large scale.
- Indivisibilities: some technologies require minimum scale.
- Learning-by-doing: productivity rises with cumulative output.



External vs. Internal Economies of Scale

External Economies of Scale

Average cost falls as the *industry* grows, not the individual firm.

Each firm remains small; market is competitive.

Sources: shared labour pool, specialised suppliers, knowledge spillovers.

Internal Economies of Scale

Average cost falls as the *firm* grows.

Large firms can undercut small ones $\Rightarrow$ market becomes imperfectly competitive (monopoly, oligopoly, or monopolistic competition).

Type	Market	Example
External	Competitive	Vancouver mining (many small firms, cluster benefits all)
Internal	Monopolistic competition	Automobile manufacturers
Internal	Oligopoly	Aircraft (Boeing, Airbus, Bombardier)

Monopolistic Competition: The Krugman Model Setup

Krugman (1980) assumptions:

- Many symmetric firms, each producing a unique variety.
- Each variety has some substitutes, but each firm has some market power.
- **Free entry and exit:** firms enter until profit = 0.
- Cost function: $TC = F + c \cdot Q$ (fixed cost F + constant MC c).
- Price: $P = c + \mu$ (cost-plus markup), where markup depends inversely on competition.

Free entry equilibrium:

$$P^* = c + \frac{F}{Q^*} = AC \quad (\text{zero profit})$$

All variables defined:

F = fixed cost (R&D, tooling, brand)

c = marginal cost (constant)

Q = output per firm

n = number of varieties

S = total market size (home + foreign)

P = price set by each firm

μ = price markup over MC

$AC = F/Q + c$ = average cost

Krugman Model: The PP and CC Curves

Two relationships:

PP curve (price P vs. number of firms n):

$$P = c + \frac{1}{b \cdot n}$$

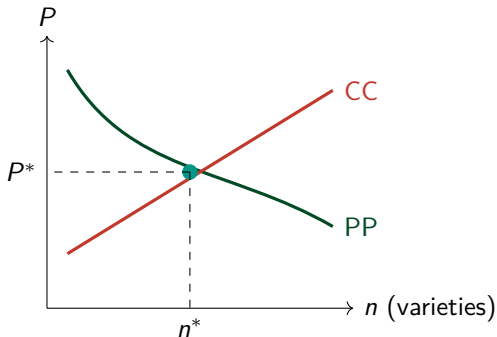
More competition $\Rightarrow$ lower markup $\Rightarrow$ lower price. *PP slopes down in n .*

CC curve (average cost = break-even price):

$$P = AC = c + \frac{n \cdot F}{S}$$

More firms $\Rightarrow$ smaller market per firm $\Rightarrow$ higher $F/Q \Rightarrow$ higher break-even price. *CC slopes up in n .*

Equilibrium: $PP = CC$.



How Trade Affects the Krugman Equilibrium

Before trade: market size = S (Home only).

After trade: market size = $S + S^*$ (Home + Foreign).

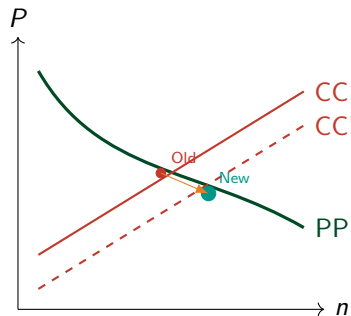
Effect on CC curve:

$$P = c + \frac{n \cdot F}{S + S^*}$$

CC shifts **down**: at each n , break-even price is lower (more output per firm amortises F).

Effect on PP curve: no change (markup depends on n only).

New equilibrium: lower P , higher n .



Three Gains from Trade Under Monopolistic Competition

1. Lower Prices

Larger market $\Rightarrow$ firms operate at larger scale $\Rightarrow$ lower $F/Q \Rightarrow$ lower AC $\Rightarrow$ lower P .
Even if each firm's technology is unchanged, more output per firm lowers unit cost.

2. More Variety

Net number of varieties available to consumers rises after trade. Each country specialises in different varieties but both countries' consumers gain access to the full range.

Consumers love variety (CES preferences).

3. Pro-Competitive

More competition from foreign varieties lowers markups. Less efficient domestic firms exit. Surviving firms are larger and more productive (Melitz, 2003).

Aggregate productivity rises.

USMCA and Canadian Consumer Electronics

Removing tariffs on US electronics increased variety, lowered prices, and forced less efficient Canadian assemblers to exit — all three gains occurred.

Measuring Intra-Industry Trade: The Grubel-Lloyd Index

Grubel-Lloyd (GL) Index

$$GL_i = 1 - \frac{|X_i - M_i|}{X_i + M_i} \in [0, 1]$$

where X_i = exports of industry i , M_i = imports of industry i .

GL = 0: pure inter-industry trade (only exports or only imports).

GL = 1: pure intra-industry trade (exports = imports).

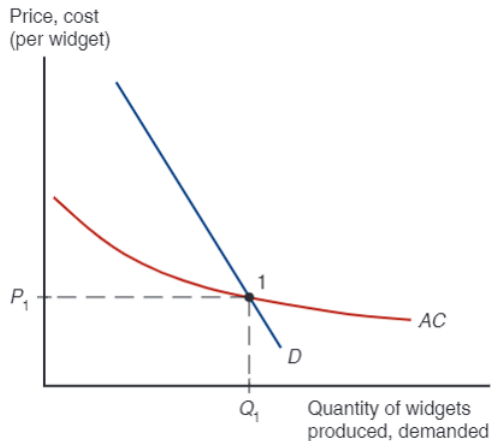
Country pair	GL index
Canada – USA	0.72
Germany – France	0.85
USA – Mexico	0.61
Japan – South Korea	0.58
USA – Bangladesh	0.12
Germany – Nigeria	0.08

Source: OECD; based on SITC 2-digit

Pattern

High GL between similar rich countries; low GL between dissimilar countries (North-South trade follows H-O, not Krugman).

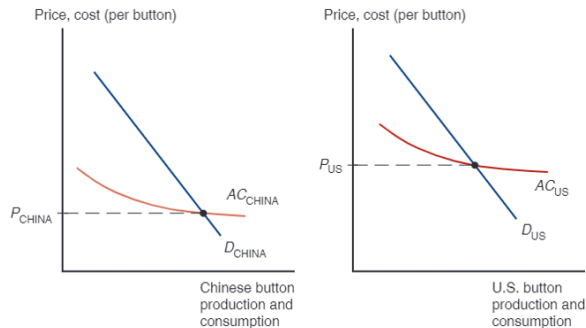
Figure 7-1: External Economies and Market Equilibrium



What this figure shows:

- **Downward-sloping AC curve:** as industry output rises, external EoS reduce per-firm costs.
- **Demand curve:** downward sloping as usual.
- **Equilibrium:** price = AC (zero profit); quantity at intersection.
- Key: the AC curve here is an *industry-level* relationship — individual firms are small and competitive, but the cluster lowers costs for all.
- This differs from internal EoS where one firm captures all cost savings.

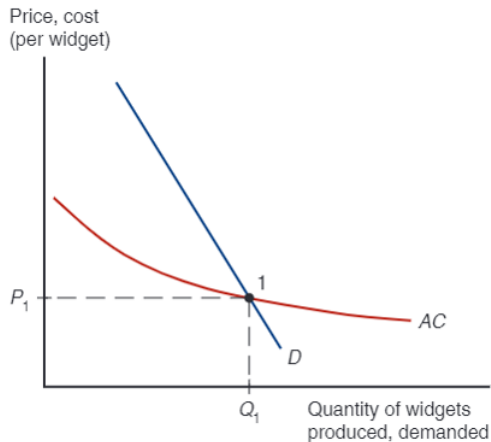
Figure 7-2: External Economies Before Trade



Reading Figure 7-2:

- Shows two countries (Home and Foreign) with *identical* external EoS AC curves.
- **Before trade:** each country produces at its domestic equilibrium.
- If Home has a *larger industry* (due to history), Home operates at a lower AC.
- **Path dependence:** the country that had a head start maintains cost advantage — even if technologies are identical.
- Trade with the larger industry may drive out the smaller one — even if the smaller could have been equally efficient.

Figure 7-3: Trade and Prices with External Economies



What this figure shows:

- After trade, the country with the **larger** industry can produce at lower cost and supplies the world market.
- The smaller industry may be **driven out** entirely (complete specialisation due to EoS, not comparative advantage).
- The world price settles at the large-country's AC curve.
- Countries can become “locked in” to comparative advantage based on historical scale, not current productivity.

Discussion

Should Canada protect its steel indus-

Industrial Clusters: Why Industries Agglomerate

Three sources of external economies (Marshall, 1890):

1. Thick labour markets:

A large pool of specialised workers reduces search costs and allows better job-worker matching. Both firms and workers benefit.

2. Specialised suppliers:

A cluster attracts specialised input suppliers (legal, financial, engineering services) that are not viable in smaller markets.

3. Knowledge spillovers:

Ideas and best practices flow more easily through face-to-face interaction. Proximity facilitates R&D collaboration.

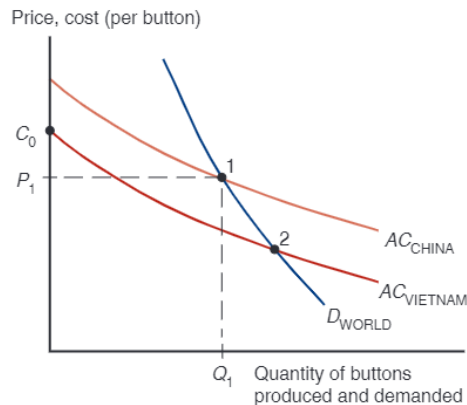
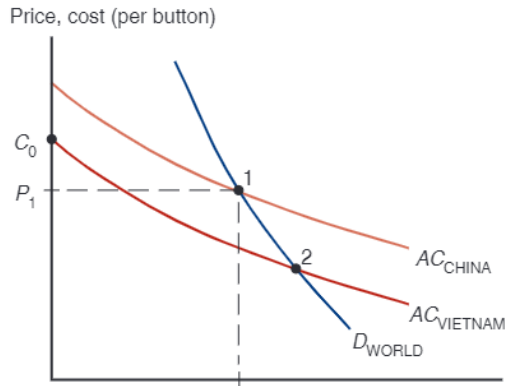


Figure 7-4: The Importance of Established Advantage

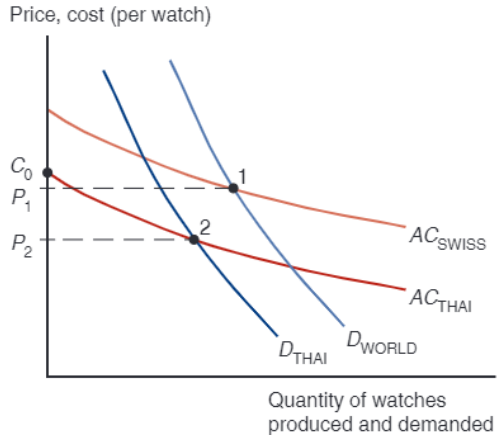


What this figure shows:

- A country that establishes a cluster early enjoys a **self-reinforcing cost advantage**.
- Even if another country could potentially achieve lower costs at the same scale, it cannot profitably enter as long as the established cluster is large.
- The established advantage *persists* over time — historical accidents (where the first firm located) determine long-run industry geography.
- This is **path dependence**: history matters for economic geography.

Canadian Example

Figure 7-5: External Economies and Potential Losses from Trade



What this figure shows:

- In some cases, trade can **harm** a country even with external EoS.
- If the foreign country has a larger cluster and lower AC, trade shifts all production there.
- The home country loses its entire industry.
- If the home industry *could have been* viable at larger scale, this is a genuine welfare loss from trade.
- This provides the strongest theoretical case for infant-industry protection — but the practical challenge remains: how do governments know which industries have this potential?

Policy Implications of Economies of Scale

The case for industrial policy based on EoS:

- External EoS create a genuine market failure: firms cannot internalise industry-wide learning benefits.
- Path dependence means historical accidents determine industry location — not efficiency.
- Temporary protection can allow a domestic industry to reach scale.

Counter-arguments:

- Governments rarely know which industries have large EoS.
- “Temporary” protection becomes permanent.

Bombardier C-Series: Vindicated Infant Industry?

Ottawa and Quebec subsidised Bombardier's C-Series aircraft development. Eventually sold to Airbus (now A220), which became globally successful.

Was the policy successful?

- The product succeeded commercially.
- But: triggered US anti-dumping case (Boeing complaint, 300% tariff).
- Bombardier had to give Airbus 50%+ stake.
- Net fiscal cost to Canadian taxpayers:

Key Takeaways

1. **EoS and product differentiation** explain intra-industry trade between similar countries.
2. **Krugman model**: trade expands market size, lowers prices (via scale), increases variety, and promotes competition — three distinct gains.
3. **Grubel-Lloyd index**: measures IIT; highest between similar rich countries.
4. **External economies** create industrial clusters with path dependence — historical accidents determine industry geography.
5. **Potential losses from trade** with EoS provide a case for industrial policy, but practical challenges are severe.
6. **Internal EoS** lead to imperfect competition — a more complex analysis covered in advanced courses.

Next: Chapter 9 — Instruments of Trade Policy

Tariffs, quotas, and their welfare effects.